

Data Scientist with expertise in Generative AI, RAG, Conversational AI, and NLP, specializing in LangChain, LangGraph, vector search, and cloud-based AI systems. Experienced in building scalable multi-agent AI platforms, MLOps pipelines, and AWS deployments using Python, Terraform, Docker, and CI/CD.

Skills & Technologies

Programming Languages & Frameworks: Python, LangChain, TensorFlow, PyTorch, Keras, FastAPI, NumPy, Pandas, scikit-learn
Machine Learning & AI: Deep Learning, NLP, Generative AI, Large Language Models (LLMs), Langchain, Retrieval-Augmented Generation (RAG), Computer Vision, EDA, Feature Engineering, vector search databases.
Cloud & MLOps: Azure, AWS (Lambda, ECS, API Gateway, S3, DynamoDB, RDS, Bedrock, SageMaker), Terraform, Docker, CI/CD,
Databases: MySQL, PostgreSQL, NoSQL (DynamoDB), **DevOps & Tools:** Git, Jira, Concourse CI/CD, Jenkins

Senior Associate Data Scientist | Publicis Sapient (2025 - Present)

Mforce (Client-Meta)

Built a **multi-agent orchestration platform** using **LangGraph**, **FastAPI**, and **CopilotKit** to guide users through AI-driven marketing experiment design and execution. Architected **three scalable microservices** with seamless **MCP tool integration**, **hybrid semantic search**, and **real-time conversational AI** capabilities.

Applied prompt engineering and few-shot learning techniques to enhance conversational AI accuracy and robustness.

Implemented advanced features including **state checkpointing** across **PostgreSQL** and **MongoDB**, and **Langfuse-backed prompt versioning** for observability and reproducibility, scalable **AWS deployment** using **ECR** and **Fargate**.

Leveraged **async/await** patterns throughout the platform to optimize performance and responsiveness, while delivering robust support for **complex multi-turn workflows**, **graceful error recovery**, and **human-in-the-loop interrupt handling**.

Analyst Programmer | Fidelity International (2022 - Present)

1. Firewall Automation System

Implemented a **Random Forest Classifier** model to classify network risk zones, improving security accuracy to 94% and created a **xgboost** model to recommend firewall deployment with a 96% accuracy rate.

Automated firewall rule generation, reducing manual effort by 80% and deployment time by 60%.

Integrated continuous monitoring & feedback loops, retraining models for adaptive security enhancements.

2. Conversational AI-Powered Search & RAG Chatbot

Developed an advanced multi-agent chatbot system leveraging LangChain and LLM for robust information retrieval.

Integrated **RAG-based Q&A** using vector databases including **FAISS** to support context-aware, document-driven conversations.

Utilized **HuggingFace embeddings** and **RecursiveCharacterTextSplitter** for efficient chunking and semantic search optimization.

Designed and deployed intelligent multi-agent workflows using **LangChain** and **agentic orchestration frameworks** like **LangGraph** and **CrewAI** to facilitate modular task delegation and dynamic tool usage.

3. CISCO-ACI Network Configuration

Built a software tool for automating CISCO-ACI switch configurations, enhancing network efficiency by 30%.

4. Automated External DNS Management

Created an automation solution for DNS record creation, reducing errors and improving accuracy by 95%.

Key Projects:

1. Stock prediction model based on time-series financial data

Developed a stock market prediction model using **LSTM-RNN**, achieving **MSE = 0.032**.

2. Emotion Detection using Deep Learning

Developed a CNN-based model for human emotion recognition, enhancing classification accuracy.

Education

Madhav Institute of Technology & Science, Gwalior (2018 - 2022)

Degree: Bachelor of Technology (B.Tech.)

CGPA: 7.84

Relevant Coursework: Machine Learning, AI, Cloud Computing, OOPS, DBMS, Data Structures & Algorithms

Certifications & Achievements:

AWS Certified Cloud Practitioner,

Qualified GATE 2022,

Machine Learning Bootcamp – iNeuron,

Generative AI with langchain and huggingface – krishai technologies private limited